DRAFT FOR DEVELOPMENT

DD ENV
13670-1:2000

Execution of concrete structures —

Part 1: Common

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ICS 91.080.40

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National foreword

This Draft for Development is the official English language version of ENV 13670-1:2000. It results from a programme of work undertaken by CEN to make available a common set of standards for execution of concrete structures.

This publication is not to be regarded as a British Standard.

It is being issued in the Draft for Development series of publications and is of a provisional nature. It should be applied on this provisional basis, so that information and experience of its practical application mat be obtained.

Comments arising from the use of this Draft for Development are requested so that UK experience can be reported to the European organization responsible for its conversion into a European Standard. A review of this publication will be initiated 2 years after its publication by the European organization so that a decision can be taken on its status at the end of its three-year life. The commencement of the review period will be notified by an announcement in *Update Standards*.

According to the replies received by the end of the review period, the responsible BSI Committee will decide whether to support the conversion into a European Standard, to extend the life of the prestandard or to withdraw it. Comments should be sent in writing to the Secretary of BSI Subcommittee B/525/2, Structural use of concrete, at 389 Chiswick High Road, London. W4 4AL, giving the document reference and clause number and proposing, where possible, an appropriate revision of the text.

A list of organizations represented on this committee can be obtained on request to its secretary.

Cross-references

The British Standards which implement international or European publications referred to in this document may be found in the BSI Standards Catalogue under the section entitled “International Standards Correspondence Index”, or by using the “Find” facility of the BSI Standards Electronic Catalogue.

Summary of pages

This document comprises a front cover, an inside front cover, the EN title page, pages 2 to 60, an inside back cover and a back cover.

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EUROPEAN PRESTANDARD

PRÉNORME EUROPÉENNE

EUROPÄISCHE VORNORM

ENV 13670-1

January 2000

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English version

Execution of concrete structures - Part 1: Common

This European Prestandard (ENV) was approved by CEN on 27 November 1999 as a prospective standard for provisional application.

The period of validity of this ENV is limited initially to three years. After two years the members of CEN will be requested to submit their comments, particularly on the question whether the ENV can be converted into a European Standard.

CEN members are required to announce the existence of this ENV in the same way as for an EN and to make the ENV available promptly at national level in an appropriate form. It is permissible to keep conflicting national standards in force (in parallel to the ENV) until the final decision about the possible conversion of the ENV into an EN is reached.

CEN members are the national standards bodies of Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and United Kingdom.



EUROPEAN COMMITTEE FOR STANDARDIZATION
COMITÉ EUROPÉEN DE NORMALISATION
EUROPÄISCHES KOMITEE FÜR NORMUNG

Central Secretariat: rue de Stassart, 36 B-1050 Brussels

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Foreword

This European Prestandard has been prepared by Technical Committee CEN/TC 104 "Concrete (performance, production, placing and compliance criteria)", the secretariat of which is held by DIN.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to announce this European Prestandard: Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and the United Kingdom.

Because of the close connection between design rules and rules for execution, CEN/TC 104/SC 2 has developed this prestandard in conjunction with CEN/TC 250/SC 2.

This prestandard supersedes the execution and tolerance clauses in ENV 1992-1 Eurocode 2: Design of concrete structures and the execution rules in prEN 206:1997: Concrete - Performance, production, placing and compliance criteria.

This prestandard specifies the required level of execution in assembling products such as fresh concrete, reinforcement, precast concrete elements etc. into a structure that achieves the intended level of mechanical resistance and stability.

This prestandard has three functions:

- to transfer the requirements set during design from the designer to the constructor i.e. to be link between design and execution;
- to give a set of standardized technical requirements for the execution when ordering a concrete structure;
- to serve as a check list for the designer to ensure that he provides the constructor with all relevant technical information for the execution of the structure (see Annex A).

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Introduction

(1) This European Prestandard assumes:

– the availability of a comprehensive design of the structure
– a project management in charge of the supervision of the works which will ensure the execution of a conforming structure;
– a site management which will take charge of the organisation of the works and ensure the correct and safe use of the equipment and machinery, the satisfactory quality of materials, the execution of a conforming structure and its safe use up to the delivery of the works.

When precast elements are used, the following additional assumptions are made:

– the availability of a specific design of the precast elements conforming to the relevant standards
– the availability of a design coordination between precast elements and site manufactured components.
– a technical specification of the precast structure with instructions for installation;
– there is an erection management to direct the erection team.

(2) This European Prestandard presupposes that the work is carried out with the necessary skill and adequate equipment and resources to perform the work in accordance with the requirements of the project specification and the requirements of this European Prestandard.

It is assumed that generally accepted rules for good workmanship for the various activities are known and applied.

(3) It is assumed that the constructor will conform to provisions valid at the construction site with respect to:

– qualifications for the personnel doing the various activities covered by this prestandard;
– health and safety aspects of construction.

(4) This prestandard assumes that the finished structure after completion is used as intended in the design and submitted to the planned inspection and maintenance necessary to achieve the intended design life and to detect weaknesses or any unexpected behaviour.

1 Scope

(1) This European Prestandard gives common requirements for execution of concrete structures. In particular, this part gives requirements for structures designed according to ENV 1992 -1 and for the concrete part of composite structures designed according to ENV 1994-1.

(2) In the case of civil engineering works, different or additional requirements need to be considered and, if required, given in the project specification.

(3) This prestandard permits the project specification to state specific requirements relevant to the particular structure.

(4) This prestandard is applicable to temporary as well as permanent concrete structures.

(5) Additional or different requirements should be considered and, if required, given in the project specification when using:

– Lightweight aggregate concrete;
– other materials (e.g.fibres) or constituent materials;
– special technologies/innovate designs.

(6) Small and simple concrete works and secondary structures of minor importance, defined as such in provisions valid at the construction site, are not within this prestandard.

(7) This prestandard does not apply to concrete members used only as equipment for the execution.

(8) This prestandard does not cover the specification, production and conformity of concrete.

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(9) This prestandard is not applicable to the production of precast concrete elements made in accordance with product standards.

(10) This prestandard does not cover the requirements for concrete members in special geotechnical works such as pile foundations, ground anchors, slurry walls, etc.

(11) This prestandard does not cover safety and health aspects of execution.

(12) This prestandard does not state requirements for quality assurance or for qualification of personnel for the various activities.

(13) This prestandard does not cover contractual issues or responsibilities for the identified actions.

2 References

(1) This European Prestandard incorporates by dated or undated reference, provisions from other publications. These references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Prestandard, only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

(2) All relevant national standards not covered by European Standards should be stated in the project specification.

2.1 Normative references

prEN 206:1997 Concrete – Part 1: Specification, performance, production and conformity

NOTE Presently EN 206 is available as prEN 206:1997 (see also clause 6.6(3), 11.2, E8.5, G 11.1 and G 11.7) and prEN 206-1 is in preparation.

EN 446:1996 Grout for prestressing tendons - Grouting procedures

EN 447:1996 Grout for prestressing tendons - Specifications for common grout

EN 523 Steel strip sheaths for prestressing tendons - Terminology - Requirements and quality control

EN 1065 Adjustable telescopic steel props - Product specifications, design and assessment by calculation and tests

ENV 1991 Eurocode 1: Basis of design and actions on structures

ENV 1992 Eurocode 2: Design of concrete structures

ENV 1994 Eurocode 4: Design of composite steel and concrete structures

prEN 10080:1999 Steel for the reinforcement of concrete - Weldable ribbed reinforcing steel Part 1 – Part 6 [1)]

ENV 10138 Steel for the prestressing of concrete. [1)]

2.2 Informative references

ISO 1803-1 Building construction – Tolerances – Vocabulary – Part 1: General terms

ISO 4463-1 Measurement methods for building – setting – out and measurement – Part 1: Planning and organization, measuring procedures, acceptance criteria

1) European Standards for reinforcement and prestressing steels (prEN 10080 and prEN 10138) are presently in preparation. Until these are issued and implemented, national standards apply.

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3 Definitions

For the purposes of this prestandard, the following definitions apply:

3.1
construction works
Everything that is constructed or results from construction operations. This term covers both building and civil engineering works. It refers to the complete construction comprising both structural and non-structural components.

3.2
constructor
The organization executing the works.

3.3
execution
All activities carried out for the physical completion of the work, i.e. scaffolding, formwork, reinforcing, concreting, curing, erection etc., and the inspection and documentation thereof.

3.4
inspection
Activities carried out in order to check that the execution is in accordance with the project specification.

3.5
method statement
Documentation describing the methods and procedures to be used to perform the work.

3.6
permitted deviation
Permitted algebraic differences between the limits of size and the corresponding reference size (see ISO 1803/1 Building construction - Tolerances-Vocabulary-Part 1: General terms).

3.7
precast concrete element
Concrete element conforming to a product standard, cast and cured in a place other than the final location of use.

NOTE In this prestandard the shorter term "precast element" is used.

3.8
project specification
Documents covering technical data and requirements for a particular project prepared to supplement and qualify the requirements of this prestandard.

3.9
reference line
Line defined in the project specification to which sizes are related.

3.10
secondary line
Any line used for the purpose of setting-out the proposed building and for checking the conformity of the building or building parts (see ISO 4463-1:1998 Measurement methods for building. Setting out measurements - Part 1: Planning and organization measuring procedures, acceptance criteria).

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3.11
surface finish
Description of the appearance of the concrete surface including aspects of geometry, texture, colour etc.

3.12
temporary structure
Structure designed for a short lifetime according to table 2.1 of ENV 1991-1.

3.13
tolerances
Permitted variation of size (see ISO 1803/1 Building construction - Tolerances-Vocabulary-Part 1: General terms).

Tolerances for precast concrete elements are subdivided as follows:

– production tolerances i.e. geometrical tolerances as defined in the product standards;
– erection tolerances i.e. geometrical tolerances relating to location, verticality, horizontality or other characteristics of the construction assembly;
– construction tolerances i.e. geometrical tolerances that are a combination of production, site construction and erection tolerances.

3.13.1
normal tolerances
The basic limits for geometrical deviations that ensures that the structure:

– satisfies the design assumptions;
– achieves other functional requirements of the construction works.

NOTE In this prestandard, normal tolerances are referred to as tolerance class 1.

3.13.2
special tolerances
More stringent tolerances than normal tolerances.

3.14
works
In the sense of this prestandard, those parts of the construction works that are structural concrete work and are described in the project specification.

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4 Documentation

4.1 Project specification

(1) It is assumed that the project specification includes:

– all necessary information and technical requirements for execution of the works and agreements made during the execution;

NOTE Annex A4, Table A1 contains a checklist of requirements and information that may have to be included in the project specification as appropriate.

– relevant European Technical Approvals and provisions valid at the construction site.

NOTE Provisions valid at the construction site are national standards and documents approved by a competent authority defined in the project specification.

(2) Before commencement of execution of any part of the works, the project specification relevant to that part of the works shall be complete and available. The following provisions should be included in the project specification:

– procedures for making alterations to previously agreed requirements;
– any requirement for the distribution, the filing and recording of technical documents used for the works.

(3) If a quality plan for the execution of the works is required, this shall be stated in the project specification.

4.2 Execution documentation

4.2.1 Quality plan

(1) If a quality control procedure is required by the project specification, it shall be available at site.

4.2.2 Special documentation

(1) If special documentation is required, the type and extent of the documentation shall be stated in the project specification.

NOTE In inspection classes 2 and 3, documentation of inspection is required, see clause 11.

5 Falsework and formwork

5.1 Basic requirements

(1) Falsework and formwork including their supports and foundations shall be designed and constructed so that they are :

– capable of resisting any action to which they are submitted during the construction process;
– stiff enough to ensure that the tolerances specified for the structure are satisfied and the integrity of the structural member is not affected.

(2) The form, function, appearance and durability of the permanent works shall not be impaired or damaged due to the performance of the falsework and formwork or their removal.

(3) Falsework and formwork shall conform to the relevant European Standard such as EN 1065.

NOTE A European standard on vertical formwork is under preparation.

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5.2 Materials

5.2.1 General

(1) Any material which leads to the fulfilment of the criteria for the structure given in 5.1 and 5.6 may be used. They shall conform to relevant product standards where they exist. The characteristics of the specific material shall be taken into account.

5.2.2 Release agents

(1) Release agents shall be selected and applied in such a way that they are not harmful to concrete, reinforcing steel, or formwork and in such a way that they have no detrimental effects on the environment.

(2) Unless specifically intended, release agents shall have no detrimental effect on the surface quality, its colour, or specified subsequent coatings.

(3) Release agents shall be applied according to the product specification or provisions valid at the construction site.

5.3 Falsework

(1) A method statement, where required, shall describe the method of erection and dismantling of temporary structures. It shall specify the requirements for handling, adjusting, intentional precambering, loading, unkeying, striking and dismantling.

(2) The design of the falsework shall take into account the deformation during and after concreting to prevent deleterious cracking in the young concrete. This may be achieved by:

– limiting the deflection and/or settlement;
– controlling the casting sequence and/or concrete specification e.g. retarding the concrete.

5.4 Formwork

(1) Formwork shall keep the concrete in its required shape until it is hardened.

(2) Formwork and the joints between boards or panels shall be sufficiently tight so as to prevent loss of fines.

(3) Formwork likely to absorb significant amounts of water from the concrete or facilitate evaporation shall be suitably wetted to reduce water uptake from the concrete, unless intended specifically for that purpose e.g. controlled permeability formwork.

(4) The internal surface of the formwork shall be clean. If the formwork is used to produce visible concrete surfaces, the treatment of the formwork surfaces shall be such that the required finish is achievable.

5.5 Special formwork

5.5.1 Slipforming

(1) When using slipforming, the design of the system shall take into account the properties of the formwork material and make provision for controlling the geometry of the works.

(2) Due to the need to batter the forms and the form friction on the young concrete, a continuous guidance system between the reinforcement and the form shall be used to ensure the required concrete cover within the tolerances given in clause 10.

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5.5.2 Other special formwork

(1) Requirements shall be given in the project specification.

5.6 Surface finish

If special finishes are required, these shall be stated in the project specification. Trial concrete panels of suitable size may be specified as a basis for approving the surface quality.

NOTE Surface finish depends on the type of formwork, concrete (aggregates, cement, addition, admixtures), execution and protection during the subsequent construction.

5.7 Inserts in formwork and embedded components

5.7.1 General

(1) Temporary inserts to keep the formwork in place, bars, ducts and similar items to be cast within the section and embedded components e.g. anchor plates, anchor bolts, spacers, shall :

– be fixed robustly enough to ensure that they will keep their prescribed position during placing and concreting;
– not introduce unacceptable actions on the structure;
– not react harmfully with the concrete, the reinforcement or prestressing steel;
– not produce unacceptable surface blemishes;
– not impair the functional performance and the durability of the structural member;
– not prevent adequate placing and compaction of the fresh concrete.

(2) Any embedded item shall have sufficient strength and stiffness to preserve its shape during the concreting operation and be free of contaminants that would affect them, the concrete or reinforcement.

5.7.2 Temporary inserts

(1) Recesses and holes used for temporary works shall be filled and finished with a material of similar quality to the surrounding concrete, unless the function of the member is such that they may remain open or another method is agreed or specified.

5.8 Removal of formwork and falsework

(1) Falsework and formwork shall not be removed until the concrete has gained sufficient strength :

– to resist damage to surfaces that may arise during the striking;
– to carry the actions imposed on the concrete member at that stage;
– to avoid deflections beyond the specified tolerances due to the elastic and inelastic (creep) behaviour of the concrete.

Striking shall be made in a manner that it will not subject the structure to impact, overload or damage.

The loads in falsework shall be released in a sequence that ensures the other falsework members are not subject to excessive loads. The stability of falsework and formwork shall be maintained when loads are released and during dismantling.

(4) The procedure for propping or re-propping when used to reduce the effects of the initial load, subsequent loading and/or to avoid excessive deflections shall be detailed in a method statement.

(5) If formwork is part of the curing system, the time of its removal shall be taken into account in accordance with the requirements of 8.5.

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6 Reinforcement

6.1 General

(1) The following clauses apply to site and fabricated reinforcement.

6.2 Materials

(1) Reinforcing steel shall conform to the European standard for reinforcing steel, prEN 10080:1999, when available and to provisions valid at the construction site.

(2) Each product shall be clearly identifiable.

(3) Anchorage devices and couplers shall conform to ENV 1992-1-1, a European Technical Approval or provisions valid at the construction site.

(4) The surface of the reinforcement shall be free from loose rust and deleterious substances which may adversely affect the steel, concrete, or the bond between them.

(5) Galvanised reinforcement shall only be used together with cement that has no detrimental effect on the bond to the galvanised reinforcement.

6.3 Bending, cutting, transport and storage of the reinforcement

(1) The cutting and bending of reinforcing steel shall conform to the project specification. The following requirements apply:

– bending shall be done at a uniform rate;
– where permitted by national standards or provisions valid at the construction site, bending of steel below temperatures of -5°C is permitted provided the procedure conforms to the given additional precautions;
– unless permitted by the project specification, bending by heating the bars is not permitted.

(2) For bending bars, the diameter of the mandrel used shall be suitable for the actual type of reinforcement and not less than the values given in table 5.1 of ENV 1992-1-1 (see annex C) or provisions valid at the construction site .

(3) For welded reinforcement and fabric bent after welding, the diameter of the mandrel used shall be suitable for the actual type of reinforcement and not less than the values given in table 5.2 of ENV 1992-1-1 (see annex C) or provisions valid at the construction site.

(4) Steel reinforcing bars, welded fabric and prefabricated reinforcement cages shall not be damaged during transporting, storing (clear of ground), handling and placing into position .

Straightening of bent bars shall be allowed only if:

– special equipment to limit local stresses is used,
– the procedure of straightening has been approved.

NOTE Permission may be given in the project specification or provisions valid at the construction site

(5) Reinforcement from coils shall not be used unless appropriate equipment is available and the straightening procedures are approved.

6.4 Welding

(1) Welding shall conform to provisions valid at the construction site.

(2) Welding is only permitted on reinforcing steel conforming to prEN 10080:1999 and reinforcing steel classified as weldable in provisisons valid at the construction site.

(3) Welding shall not be executed at, or near, bends in a bar. The limits given in table 5.2 of ENV 1992-1-1 apply.

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(4) Spot welding is permitted for the assembling of reinforcement provided that it is not restricted by provisions valid at the construction site.

6.5 Joints

(1) Bars shall be jointed, by laps, couplers or welding, in accordance with ENV 1992-1-1 provisions valid at the construction site.

6.6 Assembly and placing of the reinforcement

(1) The reinforcement shall be placed according to the project specification.

Note 1 Special attention should be given to reinforcement and cover at the location of holes of small dimensions which are not considered in the structural design.

Note 2 It is assumed that the project specification gives detailed information on the layout and spacing of bars, as well as any precautions to be taken in areas of congested reinforcement.

(2) The reinforcement shall be fixed and secured so that its final position is within the tolerances given in this prestandard. Assembly of reinforcement may be done with tie wire or spot welding (see 6.4 (2) and (4)).

(3) The specified cover to the reinforcement shall be maintained by the use of suitable chairs and spacers. Steel spacers in contact with the concrete surface are only permitted in a dry environment i.e. exposure class XO of prEN 206:1997.

(4) Requirement to cover apply to the nominal value, C_n , and apply to the surface of any reinforcement including possible assembly reinforcement.

7 Prestressing

7.1 General

(1) The following requirements apply to prestressed concrete procedures incorporating:

– bonded pre-tensioned construction;
– bonded post-tensioned construction;
– unbonded post-tensioned, internal or external construction.

NOTE 1 Special attention to safety measures is necessary.

NOTE 2 Provisions valid at the construction site (see 4.1.3) include approval documents issued by a European or national certification body.

7.2 Materials for prestressing

7.2.1 Post-tensioning systems

(1) Post-tensioning systems shall conform to European Technical Approval or provisions valid at the construction site.

(2) All parts of the post-tensioning system shall be compatible e.g. from the same prestressing system.

7.2.2 Sheaths

(1) Steel strip sheaths shall conform to EN 523.

(2) Sheaths of materials other than steel shall conform to provisions valid at the construction site.

(3) Casing sheaths for unbonded strands shall conform to the relevant product standard, if any, or to provisions valid at the construction site.

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7.2.3 Prestressing steel and substitutes

(1) The tendons and prestressing steel (wires, strands, bars) shall conform to 3.3 of ENV 1992-1-1and to provisions valid at the construction site, or ENV 10138 when available. For unbonded tendons (internal or external) the material properties shall conform to 3 of ENV 1992-1-5.

(2) Materials other than steel to be used for prestressing such as composite carbon-, glass- or aramid-fibres, shall conform to European Technical Approval or provisions valid at the construction site.

7.2.4 Anchorage elements and accessories

(1) Anchorage elements and accessories for the prestressing system shall be identified as those specified or permitted.

7.2.5 Tendon supports

(1) Tendon supports shall:

– not be deleterious to either steel or concrete;
– be stiff enough to ensure a stable fixing of the tendons in their required position;
– not damage the sheaths.

(2) The spacing of the tendon supports shall be such as to ensure the sheaths conform to the required line and level.

7.2.6 Cement-based grout

(1) Common grout for filling ducts and anchorages shall conform to EN 447:1996.

7.2.7 Grease, wax or other products.

Grease or wax for filling ducts and anchorages of unbonded tendons shall conform to ENV 1992-1-5.

7.2.8 Documentation

(1) The project specification and construction documentation related to prestressing, identification documents and approval documents on materials and/or tendons shall be available on site.

(2) When materials are delivered to site, they shall be accompanied by a delivery ticket. Materials without proper documentation shall be rejected.

(3) Delivery tickets, test reports and non-conformities shall be included in the tensioning report.

7.3 Transport and Storage

(1) Materials sensitive to corrosion e.g. prestressing steel, sheaths, anchorage devices, couplers, prefabricated tendons and tendons fabricated on site, shall be protected from harmful influences during transport and storage and also whilst placed in the structure prior to permanent protection. Materials that have corroded significantly shall be replaced by conforming materials.

(2) Cement, dry additions and admixtures for grout shall be protected from water and moisture during delivery and storage on site.

7.4 Fabrication of tendons

(1) The prestressing tendons shall be assembled in accordance with the European Technical Approval or provisions valid at the construction site.

(2) The type and class of prestressing steel shall be recorded in the inspection reports.

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(3) Welding of prestressing steel or anchorages, oxygen cutting or welding of steel in the vicinity of prestressing steel is not permitted. Welding of pressure distribution spirals, anchor plates and spot welding of perforated plates is not permitted unless otherwise stated in the project specification.

(4) Sheaths and their joints shall be sealed against the ingress of water.

7.5 Placing of the tendons

7.5.1 General

(1) Prestressing tendons shall be placed and secured in a manner that maintains their location within the permissible tolerances (see 7.2.6 (1) and 10.6).

(2) The tendons shall have a straight entry into the anchorage and the couplers.

7.5.2 Pre-tensioned tendons

(1) Unbonded lengths of the prestressing steel shall be adequately protected against corrosion.

7.5.3 Post-tensioned tendons

(1) Vents on the sheaths shall be provided at both ends, at the points of the tendon where air or water may accumulate. In the case of sheaths of considerable length, vents or inlets may be required at intermediate positions.

(2) Vents shall be properly marked to identify the cable.

(3) The sheaths and vents shall be secured to withstand the effects of placing and compacting of the concrete.

7.5.4 Internal and external unbonded tendons

(1) Unbonded tendons shall be adequately sealed against penetration of moisture.

7.6 Tensioning

7.6.1 General

(1) Tensioning shall conform to a prearranged and approved tensioning programme. Force (pressure) and elongation shall be recorded in an inspection report.

(2) Written instructions for the tensioning shall be available on site.

(3) Stressing devices shall be selected from those permitted by the prestressing system.

(4) The valid calibration records for the force measuring devices shall be available on site before the tensioning starts.

(5) Application and/or transfer of prestressing to a structure shall be done progressively and is only allowed when the concrete strength conforms to 4.2.3.5.7 of ENV 1992-1-1 and is equal to or greater than the minimum compressive strength required by the prestressing system.

NOTE Having adequate concrete strength in the anchorage areas is of particular importance.

(6) If during the stressing of post-tensioned tendons, the calculated elongation cannot be achieved within ± 5% of the specified total tensioning force or within ± 10 % of the specified tensioning force for a single tendon, action shall be taken in accordance with the the project specification.

(7) If during the stressing of pre-tensioned tendons, the calculated elongation cannot be achieved within ± 3% of the specified total tensioning force or within ± 5% of the specified tensioning force for a single tendon, action shall be taken in accordance with the project specification.

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(8) The results of the tensioning programme and its conformity or non-conformity to the requirements shall be recorded in an inspection report.

7.6.2 Pre-tensioned tendons

(1) If the fresh concrete cannot be cast in due time after tensioning, temporary protective measures shall be taken which will not affect the bond or have a detrimental effect on the steel and/or the concrete.

7.6.3 Post-tensioned tendons

(1) Tensioning is not permitted at ambient temperatures below -10°C. unless stated otherwise in provisions valid at the construction site.

(2) Tensioning shall not take place at in-situ concrete temperatures below +5°C unless the procedure conforms to special arrangements given in provisions valid at the construction site.

(3) In the case of deviation from the planned performance during tensioning, the cutting off of tendon ends or grouting is not permitted. Works which can impair the re-tensioning shall not be carried out. All these works shall be postponed until the revised tensioning report has been approved.

7.6.4 Internal and external unbonded tendons

(1) Clauses (1), (2) and (3) of 7.6.3. apply.

7.7 Protective measures (grouting, greasing, concreting)

7.7.1 General

(1) Written instructions shall be provided for the preparation and execution of the protective measures.

(2) Grouting devices shall conform to EN 446 and be selected from those permitted by the prestressing system.

(3) Results from the inspection and whether they conform to the requirements for the protection shall be recorded in the inspection report, see 11.3and annex D

(4) Anchorage areas and end caps shall be protected as well as the tendons.

7.7.2 Pre-tensioned tendons

(1) The ends of the tendons shall be protected against corrosion.

7.7.3 Post-tensioned bonded tendons

(1) Grouting of post-tensioned bonded tendons shall conform to EN 446:1996 and EN 447:1996.

7.7.4 Internal or external unbonded tendons

(1) Where external tendons are to be protected by grout, grout and grouting shall conform to 7.7.3.

(2) In other cases the sheaths and anchorages of the tendons shall be filled by the specified method with a non-corrosive grease or wax confoming to ENV 1992-1-5.

7.7.5 Grouting operations

(1) The mixing process (batching, w/c ratio, procedure, time) shall ensure the required properties by conforming to EN 446:1996 and EN 447:1996 or provisions valid at the construction site.

(2) Grouting shall conform to EN 446:1996 or provisions valid at the construction site.

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(3) Where required, e.g. large diameter of ducts, vertical or inclined ducts, post-injection shall conform to 7.8 of EN 446:1996.

(4) The volume injected shall be comparable with the theoretical free volume in the duct.

(5) Any void in the ducts shall be vacuum grouted or re-injected.

(6) In the case of vacuum-injection, the free volume in the ducts shall be measured. The amount of grout injected shall be comparable with this volume.

7.7.6 Greasing operations

(1) Greasing shall be carried out at a continuous and steady rate.

(2) The volume injected shall be comparable with the theoretical free volume in the duct. The change of grease volume with temperature shall be taken into account.

(3) After completion of greasing, unintended loss of grease from the ducts shall be prevented by sealing them under pressure.

8 Concreting

8.1 Specification of concrete

(1) Concrete shall be specified and produced in accordance with prEN 206:1997.

NOTE Check that the specification of concrete includes all the requirements for execution.

8.2 Delivery, reception and site transport of fresh concrete

Concrete shall be inspected at the point of placing.

NOTE Table G.3 gives guidance on the minimum level of inspection.

(2) Detrimental changes of the fresh concrete, such as segregation, bleeding, paste loss or any other changes shall be minimised during loading, transport and unloading as well as during conveyance on site.

(3) When required, samples for identity testing shall be taken at the point of placing or in the case of ready-mixed concrete, at the point of delivery.

8.3 Pre-concreting operations

(1) A concreting and inspection plan shall be prepared for the execution in Inspection classes 2 and 3, including all actions to be dealt with in this prestandard, relevant to the execution.

(2) When required initial testing of concreting by trial casting, shall be documented before the start of execution.

(3) All preparation works shall be completed, inspected and documented as required for the inspection class before the casting is initiated.

(4) If concrete is placed directly against ground or rock, the fresh concrete shall be protected against intermixing and water suction.

(5) Ground, rock, formwork or structural parts in contact with the section to be cast shall have a temperature which does not result in freezing of the concrete before it has sufficient strength to resist the effects of freezing.

(6) If the ambient temperature is forecasted to be below 0°C at the time of casting or in the curing period, precautions shall be planned to protect the concrete against damage due to freezing.

(7) If the ambient temperature is forecasted to be high at the time of casting or in the curing period, precautions shall be planned to protect the concrete against damaging effects.

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NOTE Examples are given in E.8.5.

8.4 Placing and compaction

(1) The concrete shall be placed and compacted in order to ensure that all reinforcement and cast-in items are properly embedded within the tolerances on the cover in compacted concrete and that the concrete achieves its intended strength and durability.

NOTE Particular care in ensuring proper compaction is required at changes in cross sections, in narrow locations, at boxouts, at congested reinforcement arrangements and at construction joints.

(2) The rate of placing and compaction shall be high enough to avoid cold joints and low enough to prevent excessive settlements or overloading of the formwork and falsework.

NOTE 1 A cold joint may form during casting if the concrete in the casting front sets before placing and compaction of the next layer of concrete.

NOTE 2 Additional requirements on the placing method and rates of placing may be needed where there are special requirements for the surface finishes.

(3) Segregation shall be minimised during placing and compaction.

(4) The concrete shall be protected against adverse solar radiation, strong wind, freezing, water, rain and snow during placing and compaction.

(5) Light-weight aggregate concrete shall not be pumped unless it is documented that pumping has no significant effect on the strength of the hardened concrete.

8.5 Curing and protection

(1) Concrete in its early life shall be cured and protected:

- to minimise plastic shrinkage;
- to ensure adequate surface strength;
- to ensure adequate surface zone durability;
- from freezing;
- from harmful vibration, impact or damage.

NOTE If high strength concrete is used, special consideration should be given to preventing plastic shrinkage cracking.

(2) Methods of curing shall achieve low evaporation rates from the concrete surface or keep the surface permanently wet.

NOTE Natural curing is sufficient when conditions throughout the required curing period are such that evaporation rates from the concrete surface are low, e.g. in damp, rainy or foggy weather.

(3) On completion of compaction and finishing operations on the concrete and where necessary, the surface shall be cured without delay. If needed to prevent plastic shrinkage cracking on free surfaces, temporary curing shall be applied prior to finishing.

(4) The duration of applied curing shall be a function of the development of the concrete properties in the surface zone.

(5) For concretes to be exposed to exposure classes X0 or XC1 conditions only (see prEN 206:1997), minimum curing periods shall be 12 hours, provided the set does not exceed 5 hours, and the surface concrete temperature is equal to or above 5°C.

Unless specified otherwise in provisions valid at the construction site, the following rules shall apply:

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– Concrete for use in classes other than XO or XC1 shall be cured until the surface concrete strength has achieved at least 50 % of the specified compressive strength.
– National standards or provisions valid at the construction site may translate these requirements into periods of time or conformity to table E.1 in annex E may be deemed to satisfy the requirements of this clause.

(6) Curing compounds are not permitted on construction joints, on surfaces to be treated or surfaces where bonding of other materials is required, unless they are fully removed prior to the subsequent operation, or they are proven to have no detrimental effects on the subsequent operations.

Unless permitted by the project specification, curing compounds shall not be used on surfaces with special requirements for the surface finish.

(7) The concrete surface temperature shall not fall below 0°C until the concrete surface has reached a strength where freezing can be resisted without damage (usually when $f_c > 5$ MPa).

(8) Unless specified otherwise in provisions valid at the construction site, the peak temperature of the concrete within a component shall not exceed 65°C, unless data are provided to prove that, with the combination of materials used, higher temperatures will have no significant adverse effect on the service performance of the concrete.

NOTE The requirements for accelerated curing by application of external or internal heat and for the cooling of sections with cast-in cooling pipes are not given in this prestandard.

8.6 Post-concreting operations

(1) After form striking, all surfaces shall be inspected in accordance with the inspection class for conformity to the requirements.

(2) The surface shall be protected from damage and disfigurement during construction.

(3) Any requirement for testing hardened in situ concrete, its frequency and the conformity criteria, shall be in accordance with the project specification.

8.7 Special execution methods

(1) Special methods shall be specified. Any changes shall be agreed and recorded in the project specification.

(2) Execution with special concrete such as light-weight concrete, high-strength concrete, heavy-weight concrete, underwater concrete etc. shall conform to provisions valid at the construction site, agreed procedures or known or proven methods.

(3) Concrete for slipforming shall have an appropriate set. Slipforming shall be performed with adequate equipment and methods ensuring that the specified cover to the reinforcement, concrete quality and surface finish are achieved.

8.8 Concreting of composite structures

(1) Concreting of composite structures shall conform to this prestandard.

9 Execution with precast concrete elements and site manufactured components

9.1 General

(1) This prestandard gives requirements for the construction operations involving structural site manufactured components or structural precast elements from their reception at the site, until the completion of their installation and final acceptance.

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(2) When site manufactured components or precast elements are used, the existence of the design coordination between them and the structural performance of the overall structure shall be verified.

9.2 Factory produced precast elements

(1) The factory produced precast elements, up to the reception of the elements at the site, are in the scope of the relevant European product standards or European technical approval. If there are no European technical specifications, provisions valid at the construction site apply.

9.3 Site manufactured components

(1) Site manufactured components may be treated as precast elements if they conform to the relevant European product standard.

(2) Site manufactured components not conforming to any product standard, shall not be considered as precast elements. Their manufacture is covered by this prestandard.

(3) The requirements for operations following the production of site manufactured components are the same as for factory produced precast elements.

9.4 Handling and storage

9.4.1 General

(1) Handling, storage and protection of the precast elements shall be carried out in accordance with the project specification.

9.4.2 Handling

(1) A lifting scheme defining the suspension points and forces, the arrangement of the lifting system and, where necessary, any special provisions shall be available.

(2) The total mass and any possible deviation shall be available for each element.

9.4.3 Storage

(1) Storage instructions for the element shall define the storage position and the permissible support points, the maximum height of the stack, the protective measures and, where necessary, any provisions required to maintain stability.

9.5 Placing and adjustment

9.5.1 General

(1) Requirements for the placing and adjustment of the precast elements shall be given in the erection specification.

(2) Before any delivery of precast elements, the erection specification necessary for their handling and possible on site storage shall be available at site.

(3) The work programme with the sequence of on-site operations shall be available at site.

(4) Erection shall not be started until the above items are satisfactorily verified.

9.5.2 Placing

(1) The erection specification shall define the arrangement of the supports, the necessary props and, where necessary, the temporary stability provisions.

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(2) Where necessary, the access and work positions shall be shown in the erection specification for the guiding of the element and the reaching of lifting devices.

(3) Construction measures are to be applied which ensure the effectiveness and stability of temporary and final supports. These measures shall minimise the risk of possible damage and inadequate performance.

NOTE Special advice may be required to ensure safe installation and to avoid accidental damage.

(4) The erection of the precast elements shall conform to the plans and the details of the assembly drawings and the operation sequence of the work programme.

(5) During installation, the correct position of the elements, the dimensional accuracy of the supports, the conditions of the joints and the overall arrangement of the structure shall be checked and any necessary adjustments made.

9.6 Jointing and completion works

9.6.1 General

(1) An inspection shall be carried out before the execution of jointing and before any completion works.

(2) The completion work shall be executed on the basis of the requirements given in the erection specification and taking climatic conditions into account.

9.6.2 In-situ works

(1) The placing of any additional reinforcement for the completion of the structure shall conform to clauses 6 and 7.

(2) The in situ concreting shall conform to clause 8.

9.6.3 Structural joints

(1) Connectors of any type shall be undamaged, correctly placed and properly executed to ensure an effective structural behaviour.

(2) Threaded and glued connections shall be executed according to the specific technology of the materials used.

(3) It is assumed that the project specification contains requirements to ensure that;

- joints have a size compatible with the sealing method;
- steel inserts of any type, used for joint connections, are properly protected against corrosion and fire by an appropriate choice of materials or covering;
- welded structural connections are made with weldable materials and are inspected.

10 Geometrical tolerances

10.1 General

(1) The completed structure shall be within the maximum allowable deviations to avoid detrimental effects in terms of

a) mechanical resistance and stability in transient and in service stages;

b) service performance during the use of the building;

c) placing compatibility for the erection of the structure and its non-structural components.

NOTE Unintended small deviations from the reference values which have no significant consequenses on the performance of the finished structure may be neglected.

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(2) This section contains the types of geometrical deviations relevant to building structures. Numerical values are given for structural tolerances, i.e. tolerances that have influence on the structural safety. Two structural tolerance classes are identified. This European Prestandard does not include permitted values for tolerance class 2. Tolerance class 2 has been defined in order to permit national values to be specified. Unless otherwise stated in the project specification, tolerance class 1 applies.

NOTE 1 Tolerance class 1, referred to as normal tolerances, achieves the design assumptions of ENV 1992 and the required level of safety. The use of tolerance class 2 should be used in conjunction with inspection class 3.

NOTE 2 The tolerances given in 10.4 to 10.6 are normative structural tolerances considered essential for the mechanical resistance and stability of structures in order to fulfil 10.1(1a) .

NOTE 3 Recommended values for other tolerances are given in annex F. These tolerances can be structural or non-structural depending on the function of the component. The project specification should state if these tolerances are applicable.

(3) The requirements of this clause relate to the completed structure. Where components are incorporated in a structure, any intermediate checking of such components shall be subordinate to the final checking of the completed structure.

(4) Any requirements for special tolerances shall be identified in the project specification and the following information shall then be given:

– any amendments to the permitted deviations given in this prestandard;
– any further type of deviation to be controlled, together with defined parameters and permitted values;
– whether these special tolerances apply to all relevant components or to particular components which are identified.

(5) Tolerances for surfaces between components where forces are intended to be transmitted by full contact bearing between the surfaces are not defined in this prestandard. Any requirements to such surfaces shall be stated in the project specification.

(6) Tolerances for components cast under water are not given in this prestandard.

(7) If a certain geometrical deviation is covered by different requirements (redundancy), the strictest tolerance is applicable.

(8) This prestandard does not give requirements for the combination of construction tolerances and structural deformations.

10.2 Reference system

(1) Tolerances of position in plane refers to the secondary lines in plane.

(2) Tolerances of position in height refers the secondary lines in height e.g. a transferred bench mark.

(3) Any requirement for the secondary lines shall be stated in the project specification.

NOTE ISO 4463-1:1998 Measurement methods for building. Setting out measurements - Part 1: Planning and organization measuring procedures, acceptance criteria gives guidance for setting-out the secondary lines.

10.3 Base supports (foundations)

(1) Base supports may be direct foundations on the ground, pile caps etc. Recommended values for the position of the centres of the base supports are given in figure F.1 in annex F.

NOTE Foundation on the ground may be either cast directly or made from precast concrete elements. Tolerance requirements to deep foundations, such as piles, slurry walls, diaphragms, special anchorages etc., are not given in this prestandard.

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10.4 Columns and walls

(1) Values for permitted structural deviations for columns and walls are given in figure 1.

NOTE Guidance for permitted deviations for the positions of columns and walls measured relative to the secondary lines is given in figure F.2 in annex F.

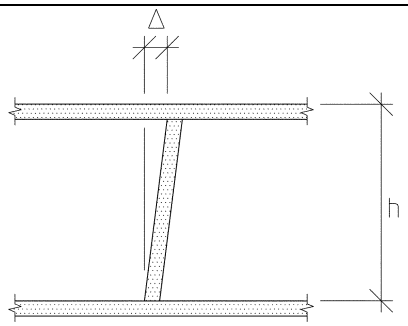
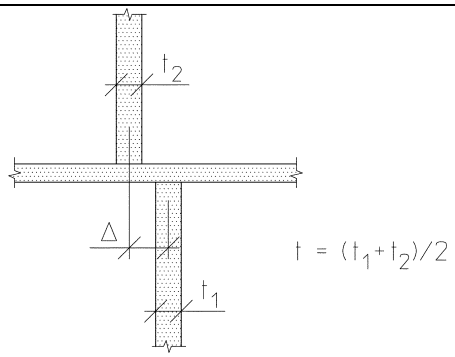
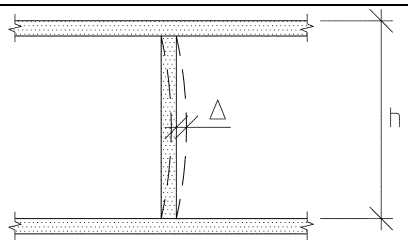
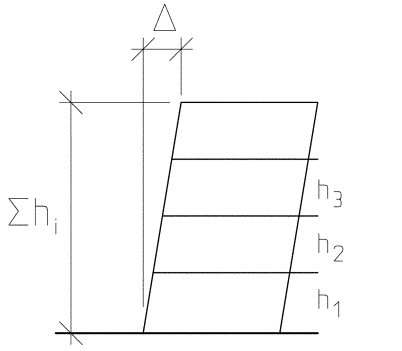
No	Type of deviation	Description	Permitted deviation △ Class 1
a		Inclination of a column at any level in a single- or a multi-storey building	The larger of h/300 or 15 mm
b		Deviation between centre lines for columns and walls	The larger of t/30 or 15 mm
c		Curvature of a column between adjacent storey levels	The larger of h/300 or 15 mm
d		Location of a column or a wall at any storey level, from a vertical line through its centre at base level in a multi-storey structure: n is the number of storeys, where n >1	The smaller of 50 mm or $\Sigma h/(200\ n^{1/2})$

Figure 1 – Permitted vertical deviations for columns and walls

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10.5 Beams and slabs

(1) The given deviations for the line and level of beams and slabs also apply to other horizontal and sloping structural components.

(2) Values for permitted structural deviations for beams and slabs are given in figure 2.

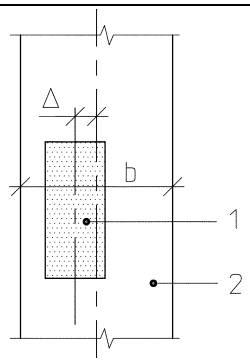
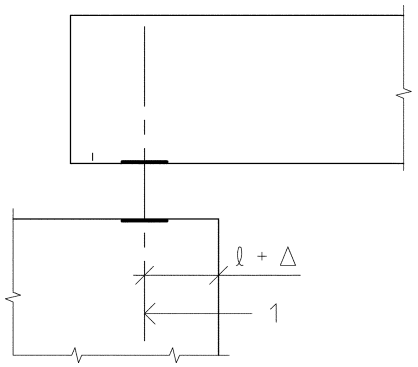
No	Type of deviation	Description	Permitted deviation △ Class 1
a	 1 Beam 2 Column	Location of a beam-to-column-connection measured relative to the column b = dimension of column in the same direction as Δ	The larger of ± b/30 or ± 20 mm
b	 1 Actual bearing axis of support	Position of bearing axis of support ℓ = Intended distance from edge	The larger of ± ℓ/20 or ± 15 mm

Figure 2 – Permitted deviations for beams and slabs

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10.6 Sections

(1) The dimensions of cross-section, the cover and position of reinforcement and prestressing reinforcement shall not deviate more from the target values than given in figure 3.

NOTE The values for deviations given do not apply to precast elements. They should conform to the relevant product standard.

(2) Conformity with the cover requirements shall be assessed for each individual reading unless provisions valid at the construction site permit a statistical approach.

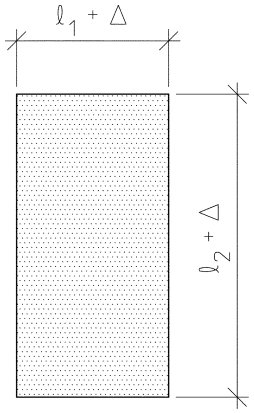
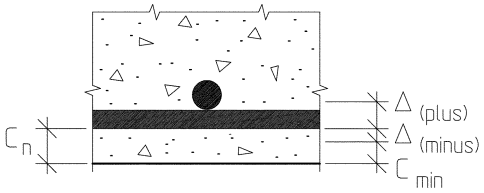
No.	Type of deviation	Description	Permitted deviation Δ Class 1
a	Cross sectional dimensions  	ℓ_i = Length of cross-sectional dimension Applicable to beams, slabs and columns For $\ell_i < 150$ mm $\ell_i = 400$ mm $\ell_i \geq 2500$ mm with linear interpolation for intermediate values	 ± 10 mm ± 15 mm ± 30 mm
	NOTE: 1. For foundations, permitted plus-deviations shall be stated in the project specification, if required. Minus-deviations are as stated. 2. Tolerances for special geotechnical concrete members cast directly onto the ground are not covered by this prestandard, e.g. slurry-walls, bored piles, etc. However, ordinary, normal foundations cast directly onto the ground are covered (i.e. blindings, etc.)		
b	Location of ordinary reinforcement Cross section  	For all values of h: Δ (minus) h ≤ 150 mm, Δ (plus) h = 400 mm, Δ (plus) h ≥ 2500 mm, Δ (plus) with linear interpolation for intermediate values	 - 10 mm + 10 mm + 15 mm + 20 mm
	c_{min} = Required minimum cover c_n = Nominal cover = c_{min} + $\lvert \Delta_{(minus)} \rvert$ c = actual cover Δ = Permitted deviation from c_n h = Height of cross-section Requirement: $c_n + \Delta_{(plus)} > c > c_n - \lvert \Delta_{(minus)} \rvert$ NOTE: Permitted plus-deviations for cover to reinforcement for foundations and concrete members in foundations may be increased by 15 mm. The given minus-deviations apply.		

Figure 3 – Permitted sectional deviations.

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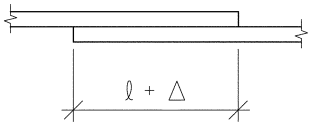
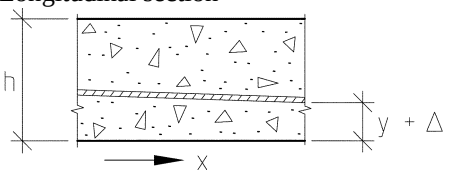
No.	Type of deviation	Description	Permitted deviation △ Class 1
c	Lap-joints 	ℓ = Lap length	- 0,06 ℓ
d	Location of prestressing reinforcement Longitudinal section 	For h ≤ 200 mm For h > 200 mm: Concrete cover measured to duct	± 0,03 h The smaller of ± 0,03 h or ± 30 mm - 15 mm
The values given apply to vertical and horizontal location. Permitted minus-deviation from nominal cover for each tendon as for reinforcement, case b.			

Figure 3 – (continued)

10.7 Surfaces and edge straightness

Recommended values for deviations for surfaces and straightness are given in figure F.5 in annex F.

10.8 Tolerances for holes and inserts

Recommended values for deviations for holes and inserts are given in figure F.6 in annex F.

11 Inspection

11.1 Inspection classes

Supervision and inspection shall ensure that the works are completed in accordance with this prestandard and the provisions of the project specification.

(2) Inspection in this context refers to verifying conformity of the properties of products and materials to be used as well as inspection of the execution of the works.

(3) Requirements for inspection shall be specified using one of the following 3 classes:

– Inspection Class 1;
– Inspection Class 2;
– Inspection Class 3.

NOTE 1 Inspection class may refer to the complete structure, to components of the structure or to certain materials/technologies used for the execution. The informative annex G gives guidance for selection of inspection class.

NOTE 2 The three inspection classes give the option to specify the required inspection based on the importance of the component/structure and the criticality of the execution for its ability to fullfill its function.

(4) The inspection class to be used shall be stated in the project specification.

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NOTE This prestandard does not deal with provisions related to the competence and degree of independence of the inspection personnel. It is assumed that any specific requirements are stated in the project specification or provisions valid at the construction site.

11.2 Inspection of materials and products

(1) The inspection of the properties of the materials and products to be used in the works is given in table 1.

Table 1 – Inspection requirements for materials and products

Subject	Inspection Class 1	Inspection Class 2	Inspection Class 3
Materials for formwork	Visual inspection.	In accordance with project specification [3)]	
Reinforcing steel	In accordance with ENV 10 080 and provisions valid at the construction site, see 11.5.1(1) [3)]		
Prestressing steel	Not applicable	In accordance with EN 10 138 or provisions valid at the construction site [3)]	
Fresh concrete; [1)] ready mixed or site mixed	In accordance with prEN 206:1997 and according to the project specification. At reception of concrete a delivery ticket shall be present. [3)]		
Other items [2)]	In accordance with the project specification [3)].		
Precast elements	In accordance with 11.8.2 [3)]		
Inspection report	Not required.	Required	

[1)] Site manufactured components are considered as components produced with "fresh concrete; ready mixed or site mixed concrete" unless they are produced according to a product standard.

[2)] For example, items such as embedded steel components etc.

[3)] Products bearing the CE-mark or third party product certification shall be checked against the delivery ticket and visually inspected. In cases of doubt, further inspection shall be undertaken to check that the product conforms to its specification. Other products shall be subject to inspection and acceptance testing as defined in the project specification

(2) If prescribed concrete is used, the relevant properties are to be checked by tests.

11.3 Scope for inspection of execution

(1) The scope of inspection to be carried out is given in table 2, unless otherwise stated in the project specification:

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Table 2 – Scope of inspection

Subject	Inspection Class 1	Inspection Class 2	Inspection Class 3
Scaffolding, formwork and falsework	Visual inspection.	Major scaffolding and formwork to be inspected before concreting, see 11.4	All scaffolding and formwork to be inspected before concreting, see 11.4
Ordinary reinforcement	Visual inspection and random measurements.	Major reinforcement to be inspected before concreting, see 11.5.1(2) and 11.5.2	All reinforcement to be inspected before concreting, see 11.5.1(2) and 11.5.2
Prestressing reinforcement	Not applicable	Components with prestressing reinforcement to be inspected before concreting, see 11.6.2 and 11.6.3	
Embedded items	Visual inspection.	According to the project specification	
Erection of precast elements	According to erection specification		
Site transport and casting of concrete	According to 11.7		
Curing and finishing of concrete	None	According to 11.7	
Stressing of prestress-ing reinforcement, including grouting	Not applicable	According to 11.6.3 and 11.6.4	
As-built geometry	Not required	According to project specification	
Documentation of inspection	Not required	As required by this prestandard	

11.4 Inspection of falsework and formwork

11.4.1 Inspection prior to concreting

(1) Before casting operations start, inspections, according to the relevant inspection class, shall include:

- geometry of formwork;
- stability of formwork and falsework and their foundations;
- tightness of formwork and its parts;
- removal of detritus (such as dust, snow and/or ice and remains of tie wire) from the section to be cast;
- treatment of the faces of the construction joints;
- removal of water from the base of the form except where special procedures for casting under water or displacing the water without intermixing are to be operated;
- preparation of the surface of the formwork;
- openings and box-outs.

11.4.2 Inspection after concreting

(1) The concrete strength shall be estimated for adequacy before removing falsework or formwork.

(2) The structure shall be checked to ensure that temporary inserts have been removed.

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11.5 Inspection of reinforcement

11.5.1 Inspection prior to concreting

(1) Before casting operations start, inspections, according to the relevant inspection class, shall confirm that:

– the reinforcement shown on the drawings is in place, and at the specified spacing;
– the cover is in accordance with the specifications;
– reinforcement is not contaminated by oil, grease, paint or other deleterious substances;
– the reinforcement is properly tied and secured against displacement during concreting;
– space between bars is sufficient to place and compact the concrete.

11.5.2 Inspection after concreting

(1) Construction joints shall be checked to ensure that the starter bars are correctly located.

11.6 Inspection of prestressing

11.6.1 Inspection for identification

(1) The identification of the materials shall be verified and checked for conformity to the specification.

11.6.2 Inspection prior to concreting

(1) Before casting operations start, inspections shall include:

– the position of the tendons, sheaths, vents, drains, anchorages, and couplers in respect of the project specification, including the concrete cover and the spacing of the tendons;
– the fixture of the tendons and sheaths, including the provision of adequate resistance against buoyancy, and the stability of their supports;
– the sheaths, vents, anchorages, couplers and their sealing are undamaged;
– the tendons, anchorages and/or couplers are not corroded;
– the cleanliness of the sheaths, anchorages, and couplers.

11.6.3 Inspection prior to tensioning

(1) The availabity on site of the documents and equipment according to the tensioning programme shall be ensured.

(2) Prior to tensioning or prior to releasing the tension force, the actual concrete strength shall be checked against the strength required.

(3) The calibration of the jacks shall be checked.

(4) When the temperature is low, the conformity to 7.6.3 shall be checked.

11.6.4 Inspection prior to grouting

(1) Before grouting starts, the inspection shall include:

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– preparation tests for grout conforming to EN 447:1996;
– ducts are open for grout through their full length and free of harmful materials, e.g. water, ice;
– vents prepared and identified;
– equipment operational;
– materials are batched and sufficient to allow for overflow;
– the results of any trial grouting on representative ducts.

(2) During grouting, the inspection shall include:

– conformity of the fresh grout tests (fluidity, segregation), see EN 447:1996;
– the characteristics of the equipment and of the grout;
– the actual pressures during grouting;
– order of blowing and washing operations;
– precautions taken to keep ducts clear;
– order of grouting operations;
– actions in the event of incidents and harmful climatic conditions;
– the location and details of any re-injection.

11.7 Inspection of the concreting operations

(1) The inspection and testing of concreting operations shall be planned, performed and documented in accordance with the inspection class, see table 3.

(2) Basic inspection is the continual inspection of conformity and normal good practice.

Table 3 – Requirements for planning, inspection and documentation

Subject	Inspection Class 1	Inspection Class 2	Inspection Class 3
Planning for inspection		Inspection plan, procedures and instructions as specified Actions in the event of a non-conformity.	Inspection plan, procedures and instructions as specified. Actions in the event of a non-conformity.
Inspection	Basic inspection.	Basic and random detailed inspection.	Detailed inspection of each casting.
Documentation	Records from all unusual events. All non-conformities and corrective action reports.	All planning documents. Records from all inspections. All non-conformities and corrective action reports.	All planning documents. Records from all inspections. All non-conformities and corrective action reports.

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11.8 Inspection of precast concrete elements

11.8.1 General

(1) Before installation, the existence of adequate site conditions shall be verified by an initial inspection.

NOTE Recommendations for the initial site inspection are given in G.11.8.1 of annex G.

11.8.2 Reception checks

(1) An initial visual inspection of the precast elements shall be carried out before unloading.

(2) As soon as possible after delivery, the precast elements shall be inspected for acceptance.

NOTE Guidance on inspection is given in G.11.8 of annex G.

11.9 Action in the event of a non-conformity

(1) Where inspection reveals a non-conformity, appropriate action shall be taken to ensure that the structure remains fit for its intended purpose.

NOTE See G.11.9 for further guidance.

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Annex A (informative): Guidance on documentation

Main clause numbers mirror those in Section 4: Documentation

A.4.1 Project specification

(1) The technical documentation for concrete structures comprises:
design calculations for individual members and the complete structure;
a project specification.

(2) The project specification comprises:

(a) Construction drawings, giving all necessary information such as geometry of the structure, amount and position of reinforcing and prestressing steel and for precast concrete elements, lifting devices, weights, inserts, etc.

(b) Description of all products to be used with any requirement for the application of the products. This information should be given on the drawings and/or in the work description.

(c) Work description which is the document that describes the inspection classes to be applied, any special tolerance, requirements for the properties of surface finish etc. The work description should also include all requirements for execution of the work, i.e. sequence of operations, temporary supports, work procedures etc.

(d) Where relevant, an erection specification for precast concrete elements.

(3) An erection specification for precast concrete elements comprises:

- installation drawings consisting of plans and sections showing the positions and the connections of the elements in the completed works;
- installation data with the required in-situ material properties and inspections;
- installation instructions with the necessary data for the handling, storing, setting, adjusting, connection and completion works (see 9.4, 9.5 and 9.6).

(4) Table A 1 gives a summary of the information that should be included into the project specification, as relevant, to be in accordance with this prestandard.

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Table A 1 – Checklist of information for inclusion in the project specification

Section	Clause	Text
1 Scope	1 (2)	Specify any additional requirements for civil engineering works
	1 (3)	Specify specific project requirements
	1 (7)	State any requirements on concrete members used as equipment for the execution
	1 (10)	State any requirements for special geotechnical works
	1 (11)	Specify requirements related to health and safety
	1 (12)	Specify requirements related to qualifications of personnel
	1 (13)	Define responsibilities, where appropriate
2. Normative references	2 (2)	Add all relevant national standards or provisions valid at the construction site
3 Definitions	3.9	Define the reference line for setting out
4 Documentation	4.1 (1)	All necessary technical information to be set out in the project specification
	4.1 (1)	National provisions which need to be respected
	4.1 (2)	Include procedure for altering project specification
	4.1 (2)	Requirements for document distribution
	4.1 (3)	State if quality plan is required
	4.2.2 (1)	State if execution documentation is required
5 Falsework and formwork		
	5.2 (2)	Specify provisions for application of release agents
	5.3(1)	Specify construction programme and Method Statement, if required
	5.4 (4)	Specify any requirements for surface finish
	5.5.2	Specify any requirements for special formwork
	5.6 (1)	Specify any requirements for special finishes or trial panels
	5.7.2 (1)	Requirements for filling temporary works inserts
6 Reinforcement	6.2 (1)	Specify types of reinforcement
	6.2 (1)	Specify provisions for reinforcing steel
	6.2 (3)	Specify permitted types of anchorages or couplers
	6.2 (3)	Specify provisions for anchorage devices and couplers
	6.3 (1)	Provide cutting and bending schedules or identify that this is a task for the constructor
	6.3 (1)	Is bending at temperatures below –5°C permitted and if so specify the precautions to be taken
	6.3 (1)	State if bending by heating in permitted
	6.3 (4)	Specify any requirements to straighten bent bars
	6.3 (5)	Procedure for straightening coils
	6.4.1	Provisions for welding of reinforcement
	6.4 (4)	Confirm that spot welding is permitted
	6.4 (5)	State if welding of galvanised or epoxy coated bars is permitted. If so, specify the repair method.
	6.5 (1)	Provisions for joints of reinforcement

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	6.6 (1)	Location of reinforcements including the position of joints
	6.6 (1)	Detail drawings for congested reinforcement
7 Prestressing	7.1 NOTE 2	Documents approved by a competent authority
	7.2.1 (1)	Approval of post-tensioning system
	7.2.2 (3)	Provisions for casing sheaths
	7.2.3 (2)	State if alternatives to prestressing steel are permitted. If so, state which types and qualities
	7.2.5 (2)	Description of tendon support
	7.4 (1)	Provisions for assembling of prestressing tendons
	7.4 (3)	Specify if welding of pressure distribution spirals, anchor plates and spot welding of perforated plates is permitted
	7.5.1 (1)	Locations of prestressing tendons
	7.6.1 (6)	Actions to be taken when accuracy of elongation or tensioning force cannot be achieved
	7.6.3 (1)	State if tensioning below –10°C is permitted. If so specify any additional precautions
	7.6.3 (1)	Provisions for tensioning at ambient temperatures below –10°C
	7.6.3 (2)	State any special arrangements for tensioning at in-situ temperatures below +5°C
	7.7.5 (1)	Provisions for the mixing process of grouting if not conforming to EN 446:1996 and EN 447:1996
	7.7.5 (2)	Provisions for the grouting if not conforming to EN 446:1996
	7.7.5 (3)	State if post-injection is required
8 Concreting	8.1 (1)	Check that all the required concrete properties have been specified
	8.3 (2)	State if a trial casting is required
	8.4 (2)	State any additional limits for the concrete surfaces
	8.5 (5)	State if different curing periods are needed
	8.5 (5)	Provisions for translating percentage of compressive strenght to periods of time
	8.5 (6)	State any restrictions for curing compounds on concrete surfaces
	8.5 (8)	State any requirements for accelerated curing or cooling
	8.5 (8)	State if higher peak temperatures are acceptable
	8.6 (3)	State any requirements to in-situ testing of hardened concrete. If so, specify the method of test, its frequency and conformity criteria
	8.7 (1)	Specify any special execution methods
	8.7 (2)	State any need to specify procedures for execution with special concretes
9 Execution with precast concrete elements	9.3 (1)	State if site manufactured components shall conform to a product standard. If so, specify the standard.
	9.4.1	Specify handling , storage and protection
	9.5 (1)	Requirements relating to placing and adjustment of precast elements
	9.6	In situ works required for completion
	9.6.2 (1)	Detailing of in situ works
	9.6.2 (3)	Specify appropriate materials or coverings
	9.6.3 (1)	Detailing of structural joints

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	9.6.3 (2)	Specify acceptable specific technologies
	9.6.3 (3)	Specify requirements to joints, inserts for joint connections and welded structural connections
10 Geometrical tolerances	10.1 (1)	State any requirements related to aesthetics
	10.1 (2)	Specify if tolerance class 2 applies
	10.1 (2)	Specify of any of I/N tolerances apply
	10.1 (4)	Specify any special tolerances and the elements to which they apply
	10.1 (5)	State any requirements for surfaces with full contact bearing
	10.1 (6)	If sections are to be cast under water, specify tolerances
	10.2 (3)	State any requirements for the secondary lines
	10.3 (1)	Check need to specify tolerances for deep foundations
	10.6 (2)	Specify permitted plus-deviations for foundations
	10.6 (2)	State if a statistical evaluation of concrete cover is permitted
11 Inspection, correction	11.1 (4)	Specify inspection class and define who is responsible for the inspection
	11.1 (4) NOTE:	Specify provisions related to inspection personnel
	11.2 Table 1	Define inspections and acceptance testing of products without a CE-marking or third party certification
	11.2 (1) Table 1	Specify provisions relating to the inspection of materials and products
	11.3 Table 2	Check if the scopes of these inspections are adequate. If not give additional requirements
	11.7 (1)	Inspection class for concreting operations
	11.9	If required specify more detailed actions in the event of a non-conformity
Annex A Documentation	A 4.1 (2)	Check if the project specification is complete
	A 4.1 (3)	If erection specification is required, check if it is complete
Annex B Falsework and formwork	B 5.5.1 (1)	Check whether the specified cover is to include allowance for surface irregularities
Annex C Reinforcement	C.6.3 (5)	Provisions for rebending at the same point
Annex D Prestressing	D 7.7.3 (1)	Check that the construction periods are adequate
Annex E Concreting	E 8.3 (6)	Specify the ambient temperature above which precautions should be planned if not in national standards or provisions valid at the construction site
	E 8.4 (9)	Specify any requirements for direct surface finishes
	E 8.5 (1)	Specify any additional requirements for the curing and protection
	E 8.5 (2) NOTE	Provisions for curing compound
	E 8.5 (8)	State any need to increase the curing periods
	E 8.5 (8)	Provisions for curing periods where sevoro conditions occur
Annex F Tolerances	F 10.7 Fig. F.6	State permitted deviation for holes and inserts
Annex G Inspection	G 11.7 (1)	State if an inspection plan is needed
	G 11.7 (1)	State who has authority to accept the form and content of inspection documentation

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	G 11.7 (2) Table G.3	Requirements to identy testing , air content and to time of arrival/placing and to temperature
	G 11.7 (2) Table G.5	Specify inspection of free surfaces
	G 11.7 (2) Table G.6	Specify planning for inspection
	G 11.7 (2) Table G.7	Specify if check with covermeter is required
	G 11.8.3 (1)	State if any additional checks are required
	G 11.9 (2)	State any required procedure for rectification in case of non - conformity

A.4.2 Execution documentation

(1) The following subjects should be considered for inclusion in the execution documentation:

- Sources of materials, material test certificates and/or suppliers' attestation of conformity;
- applications for variations and responses;
- as-built drawings or sufficient information to enable as-built drawings to be produced for the entire structure including any precast elements;
- a description of non-conformities and where applicable, the corrective actions taken;
- a record of accepted changes to the project specification;
- records of any dimensional checks at hand-over;
- a diary where the events of the construction process are reported;
- documentation of the inspections .

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Annex B (informative) : Guidance on falsework and formwork

Main clause numbers mirror those in Section 5: Falsework and formwork

B.5.1 Basic Requirements

(1) The principal actions to be taken into account in the design include the governing combinations of:

– deadloads of formwork, reinforcement and concrete;
– pressure on formwork taking into account concrete type (including possible uplift);
– construction loads (crew, equipment, etc...), including static and dynamic effects of placing, compacting and construction traffic;
– wind and snow loads.

(2) The provision of adequate bracing and its means of connection are important.

B.5.3 Falsework

(1) Wedges for the correct adjustment of falsework supports must be properly secured against slip during concreting.

(2) When propping off the ground, the influence of differential settlements should be taken into account.

B.5.4 Formwork

(1) A closable window (opening) at the bottom of the form can be helpful when cleaning out the forms.

B.5.5 Special formwork

B.5.5.1 Slipforming

Measures to secure the protection of the reinforcement may be:

– treatment of the surfaces after leaving the form e.g. floating;
– application of a suitable compound on the surfaces;
– increase of the nominal cover.

B.5.5.2 Permeable formwork lining

(1) The use of a permeable formwork lining improves the quality of the concrete in the cover zone and significantly reduces the number and size of blowholes.

B.5.7 Inserts in formwork

(1) When aluminium or galvanized steel inserts are to be used, special measures should be taken to avoid chemical reactions between the metal and the concrete.

(2) Metallic materials of different electrical potential should not be electrically connected.

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Annex C (informative): Guidance rules on reinforcement

Main clause numbers mirror those in Section 6: Reinforcement

C.6.3 Bending, cutting, transport and storage of the reinforcement

(1) Measures should be taken to avoid:

– mechanical damage (e.g. notches or dents);
– rupture of welds;
– reduction of the section through corrosion.

(2) Mechanical cutting is an appropriate method.

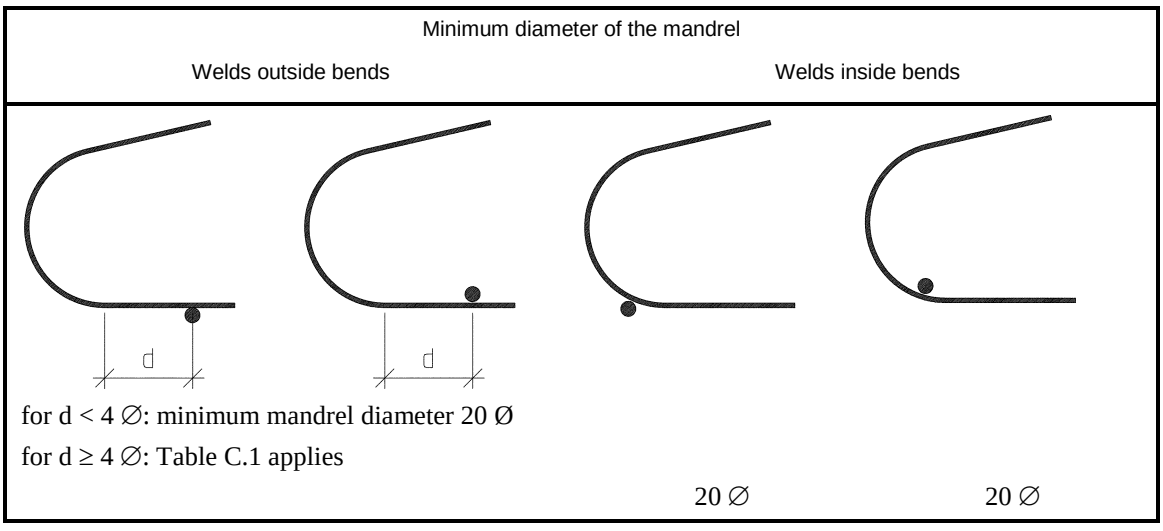
(3) Table C.1 is a reproduction of table 5.1 of ENV 1992-1-1.

Table C.1 – Minimum diameter of the mandrel

	Hooks, bends, loops		Bent-up bars or other curved bars		
	Bar diameter		Value of minimum concrete cover, perpendicular to plane of curvature		
	Ø < 20 mm	Ø ≥ 20 mm	> 100 mm and > 7 Ø	> 50 mm and > 3 ∅	≤ 50 mm and ≤ 3 Ø
Plain bars S 220	2,5 Ø	5 Ø	10 Ø	10 Ø	15 Ø
High bond bars S400, S 500	4 Ø	7 Ø	10 Ø	15 Ø	20 Ø

(4) Table C.2 is a reproduction of table 5.2 of ENV 1992-1-1.

Table C.2 – Minimum diameter of the mandrel for welded reinforcement and fabric



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(5) The following conditions should be satisfied when cold bending reinforcing steel:

– provisions valid at the construction site will state if rebending at the same point is permitted;
– boxes used to cover reinforcing bars for later connection should be designed not to adversely affect the load bearing capacity of the concrete section or the corrosion protection of the reinforcement.

C.6.4 Welding

(1) The welding methods permitted include:

– arc-welding;
– gas-shielded welding;
– flash-welding;
– spot welding.

(2) For structurally important welds, the welder should have appropriate welding certificates.

C.6.6 Assembly and placing of the reinforcement

(1) In the selection of suitable chairs and spacers as required by 6.6.(3), consideration of the loading during placing of the reinforcement and casting of the concrete should be taken into account.

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Annex D (informative): Guidance on prestressing

Main clause numbers mirror those in Section 7: Prestressing

D.7.3 Transport and storage

(1) Prestressing steel, anchorages, couplers, and ready-made tendons should be transported on carriages which are clean and free from chemical substances aggressive to the steel. Any contact with detrimental substances should be avoided by special packing in the mill or by supporting the steel in a way that prevents it coming into contact with the carriage surfaces.

(2) Transport by water should not be allowed without suitable packing.

(3) The bar diameters that can be transported and stored as coils is to be approved.

(4) Prestressing steel should not be stored in contact with the ground nor exposed to rain. Prestressing steel should preferably be stored in closed rooms at a relative humidity of less than 60 per cent.

(5) Ready-made tendons within sheaths should be protected at their ends against the penetration of moisture, preferably against condensation, and supported at distances which does not impair the stability and tightness of the sheaths.

D.7.4 Fabrication of the tendons

(1) Sockets and other connections should meet the same requirements as the sheaths.

(2) Tapes for sealing of the sheaths should be free of chloride.

(3) Prestressing steel may be cut with a disk cutter.

D.7.5 Placing of the tendons

D.7.5.3. Post-tensioned tendons

(1) Resistance against buckling of sheaths can be achieved by using a sufficiently stiff sheath or with temporary support from a polythene tube or similar.

D.7.6. Tensioning

D.7.6.1 General

(1) Tensioning is a complex operation working with high forces on the jacks and the prestressing tendons. It is an operation which requires suitable safety measures and supervision by experienced personnel.

D.7.6.2 Pre-tensioned tendons

(1) In addition to the requirements in 7.6.1 and 7.6.2, the tensioning programme should specify :

- any special sequence of tensioning;
- the jack pressure and its equivalent jack force to be attained;
- the minimum and maximum permissible tension in the tendons and their slip in the anchorages;
- the required concrete strength at the time of releasing the prestressing force.

(2) The operational suitability of reusable anchorage components should be proven by a check.

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D.7.6.3 Post-tensioned tendons

(1) In addition to the requirements in 7.6.1 and 7.6.3, the tensioning programme should specify :

– the prestressing system to be used;
– the type and grade of prestressing steel;
– the number of bars, wires or strands in each tendon;
– the required concrete strength for the application of tension;
– the order in which successive tendons have to be tensioned and any requirements for phased tensioning on a tendon;
– the calculated tensioning and jacking force as well as the elongation of the tendons;
– the anticipated slip at the anchorage;
– any necessary partial or full release of the falsework .

(2) The following should be recorded :

– the verification of the required concrete strength for the tensioning;
– the type of prestressing jack used;
– the measured jack force and the elongation of the tendon in each stage of tensioning;
– the observed slip;
– any severe deviation from the calculated tensioning force or elongation;
– if specified, the release of falsework.

D.7.6.4 Internal and external unbonded tendons

(1) The clauses (1) and (2) of D.7.6.3 apply.

D.7.7 Protective measures

D.7.7.3 Post-tensioned bonded tendons

(1) If the penetration of water or excessive humidity can be prevented, and if provisions valid at the construction site do not specify otherwise, the following construction periods are recommended :

– maximum of 12 weeks between fabrication of tendons and grouting;
– maximum of 4 weeks in the formwork before casting of concrete;
– approximately 2 weeks in the tensioned condition before applying the protective measures.

(4) If the period above between tensioning and grouting is exceeded, temporary protection should be maintained by an approved method. At appropriate intervals, flushing of the ducts with dried air or nitrogen can provide a suitable means of protection.

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Annex E (informative): Guidance on concreting

Main clause numbers mirror those in Section 8: Concreting

E.8.2 Delivery, reception and site transport of fresh concrete

(1) The receiving inspection should comprise a check of the delivery ticket prior to discharge.

(2) The receiving inspection should be documented by signing the delivery ticket when relevant.

(3) The concrete should be visually inspected during unloading. Unloading should be stopped if the appearance, judged by experience, is abnormal.

E.8.3 Pre-concreting operations

(1) Construction joints should be clean, free of laitance and wetted to a damp condition.

(2) The form should be free of detritus, ice, snow and standing water.

(3) Structural elements should be isolated from the ground by a blinding layer of at least 50 mm unless the concrete cover to the reinforcement is increased accordingly.

(4) Concreting onto frozen ground should not be permitted, unless special procedures are followed.

(5) The surface temperature at the construction joint should be above 0 °C at the time of concreting.

(6) Provisions valid at the construction site should define ambient temperatures above which precautions have to be planned to protect the concrete against damaging effects.

E.8.4 Placing and compaction

(1) Compaction should be performed by internal vibration, unless otherwise agreed.
(2) Concrete should be placed as near as practical to its final location. Vibration should be used to compact the concrete and not as a means of moving the concrete long distances.

(3) Vibration by poker or surface vibrator should be applied systematically after placing until the expulsion of entrapped air has practically ceased. Excessive vibration which might promote weak surface layers or segregation should be avoided.

(4) Normally the thickness of the concrete layer placed should be less than the height of the poker vibrator. Vibration should be systematic and include re-vibration of the top of the previous layer.

(5) Where permanent formwork is incorporated in the structure, its energy absorption should be taken into account when deciding the method of compaction and consistence of the concrete.

(6) In deep sections, re-compaction of the surface layer is recommended to compensate for plastic settlement below horizontal top reinforcement.

(7) Where only surface vibrators are used, the layer of concrete after compaction should, in normal situations, not exceed 100 mm unless proved acceptable by trial castings. Additional vibration near the supports may be required to obtain adequate compaction.

(8) Surface finishing by screeding, trowelling or floating should be carried out in a manner and at the necessary time to achieve the specified surface finish.

(9) Surface finishing should not result in laitance.

(10) Water, cement, surface hardeners or other materials should not be added during the finishing operations unless specified or agreed.

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E.8.5 Curing and protection

(1) The project specification may contain additional requirements for:

– maximum temperature difference across the cast section;
– maximum temperature difference between the cast section and previously cast sections or other forms of restraint;
– aggregate type;
– monitoring during construction.

(2) The following methods are suitable for curing used separately or in sequence:

– keeping the formwork in place;
– covering the concrete surface with vapour-proof sheets which are secured at the edges and joints to prevent draughts;
– placing of wet coverings on the surface and protection of these coverings against drying out;
– keeping the concrete surface visibly wet with suitable water;
– application of a curing compound of established suitability.

NOTE Until a European standard on curing compound is published, suitability may be based on provisions valid at the construction site.

Other curing methods of equal effectiveness may be used.

(3) The development of properties in the surface zone should be based on one of the following relationships:

– compressive strength to maturity;
– heat evolution to the total heat generated under adiabatic conditions.

(4) Table E.1 gives the duration of curing in number of days deemed to satisfy 8.5 (5).

Table E.1 – Minimum curing period for prEN 206:1997 exposure classes other than X0 and XC1

Surface concrete temperature (t), °C	Minimum curing period, days [1), 2)]			
	Concrete strength development [4)] (f_{cm2}/f_{cm28}) = r			
	rapid r ≥ 0,50	medium r = 0,30	slow r = 0,15	very slow r < 0,15
t ≥ 25	1,0	1,5	2,0	3,0
25 > t ≥ 15	1,0	2,0	3,0	5
15 > t ≥ 10	2,0	4,0	7	10
10 > t ≥ 5 [3)]	3,0	6	10	15

NOTE:

1. Plus any period of set exceeding 5 hours.
2. Linear interpolation between values in the rows is acceptable.
3. For temperatures below 5°C, the duration should be extended for a period equal to the time below 5°C.
4. The concrete strength development is the ratio of the mean compressive strength after 2 days to the mean compressive strength after 28 days determined from initial tests or based on known performance of concrete of comparable composition (see prEN 206:1997).

(5) If heat evolution is used to measure the development of concrete properties, heat ratio corresponding to the strength ratio given in this prestandard should be established by the national standardisation body.

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(6) Detailed estimates of the development of concrete properties may be based on one of the following methods:

– maturity calculation from temperature measurements taken at a maximum depth of 10 mm below the surface;
– maturity calculation based on the daily average air temperature;
– temperature-matched curing;
– other methods of established suitability.

(7) Maturity calculations should be based on an appropriate maturity function, proven for the type of cement or combination of cement and addition in use.

(8) Where specified in national standards or provisions valid at the construction site, the curing periods for concrete surfaces exposed to abrasion or other severe exposure conditions are to be increased to achieve the specified higher strength ratios.

(9) Curing compounds may penetrate the surface and make removal very difficult, therefore grit blasting or high pressure water jetting will normally be necessary if they have to be removed.

(10) The use of a curing compound containing a fugitive dye makes verification of application simple.

(11) Possible adverse effects of high concrete temperatures during curing include:

– significant reductions of strength;
– significant increase in porosity;
– delayed ettringite formation;
– increase in the temperature difference between the cast element and the previous cast restraining element.

E.8.7 Special execution methods

(1) Special methods of execution should be described and documented in a method statement, a work procedure or a work instruction.

(2) Slipforming should be controlled by a specialist. Special care should be taken to control the rate of slipforming taking account of the actual stiffening time of the concrete.

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Annex F (informative): Guidance on geometrical tolerances

Main clause numbers mirror those in Section 10: Geometrical tolerances

(1) In this annex, guidance is are given for permitted geometrical deviations. These are tolerances for geometrical quantities which are considered to have small structural influence.

F 10.3 Base supports (foundations)

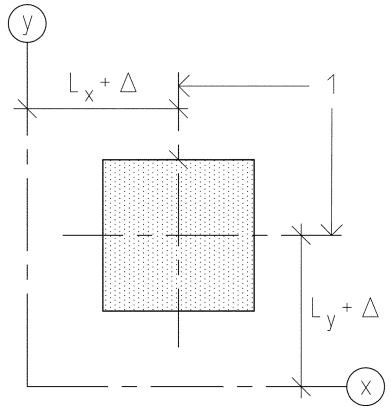
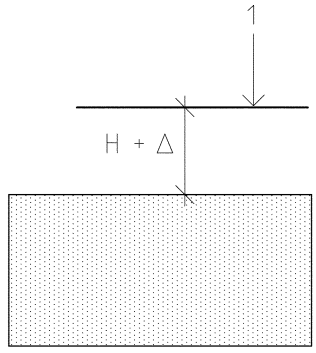
No	Type of deviation	Description	Permitted deviation △ Class 1
a	Horizontal section:  1 Support centre lines y Secondary line in y - direction x Secondary line in x - direction	Position in plan of a base support relative to the secondary lines	± 25 mm
b	Vertical section:  1 Secondary level H intended distance	Position in vertical direction of a base support relative to the secondary level	± 20 mm

Figure F.1 – Permitted deviations for the position of base supports (foundations)

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F 10.4 Columns and walls

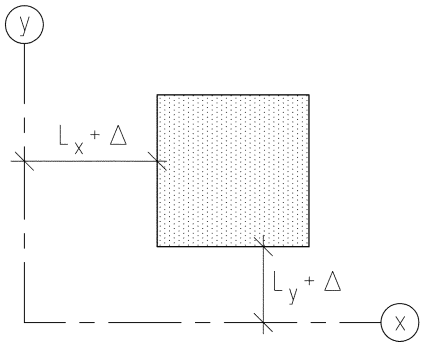
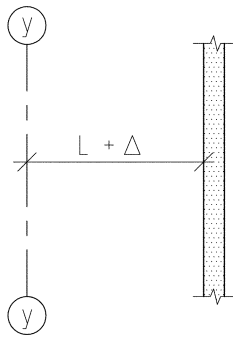
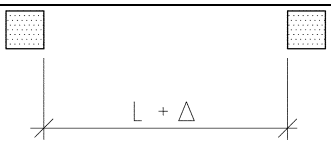
No	Type of deviation	Description	Permitted deviation △ Class 1
a	Secondary line  Secondary line	Position in plane of a column relative to the secondary lines	± 25 mm
b	Secondary line 	Position in plane of a wall relative to the secondary line	± 25 mm
c		Free space between adjacent columns or walls	The larger of ± 25 mm or ± L/600

Figure F.2 – Permitted deviations for position of columns and walls, horizontal sections

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F 10.5 Beams and slabs

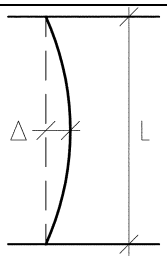
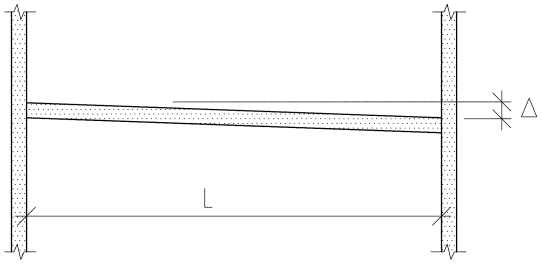
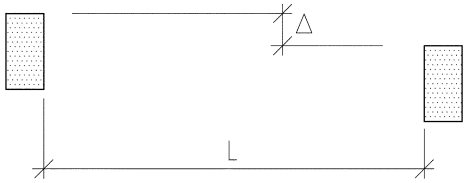
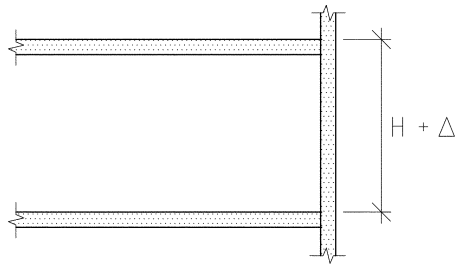
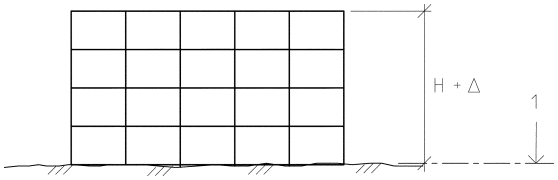
No	Type of deviation	Description	Permitted deviation △ Class 1
a		Horizontal straightness of beams	The larger of ± L/600 or ± 20 mm
b		Distance between adjacent beams, measured at corresponding points	The larger of ± L/500 or ± 15 mm, but not more than 40 mm
c		Inclination of a beam or a slab	± (10 + L/500) mm
d		Level of adjacent beams, measured at corresponding points	± (10 + L/500) mm
e		Levels of adjacent floors at supports	± 15 mm
f		Level of upper floor measured relative to the secondary system H ≤ 20 m 20 m < H < 100 m H ≥ 100 m	 ± 20 ± 0,5 (H + 20) ± 0,2 (H + 200)

Figure F.3 – Permitted deviations for beams and slabs

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F 10.6 Sections

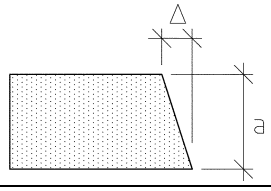
No.	Type of deviation	Description	Permitted deviation △ Class 1
a	Orthogonality of a cross section 	a = Length of cross-sectional dimension	The larger of 0,04 a or 10 mm, but not more than 20 mm. Plus- or minus-deviation

Figure F.4 – Permitted cross-sectional deviations

F 10.7 Surface straightness

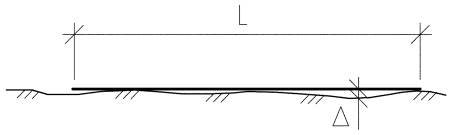
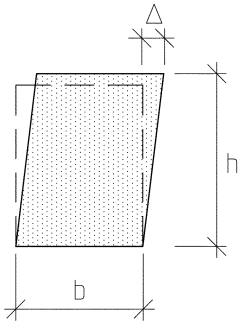
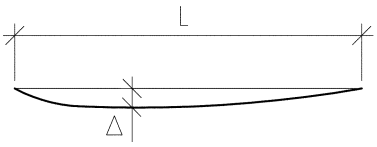
No.	Type of deviation	Description	Permitted deviation △ Class 1
a	Flatness Moulded or smoothed surface: global local Not moulded surface: global local 	 L = 2,0 m L = 0,2 m L = 2,0 m L = 0,2 m	 9 mm 4 mm 15 mm 6 mm
b	Skewness of cross-section 		The greater of h/25 or b/25, but not more than 30 mm. Plus- or minus-deviation
c	Edge straightness 	for lengths < ± 1 m for lengths >1 m	8 mm 8 mm/m, but not more than 20 mm

Figure F.5 – Permitted deviations for surfaces and edges

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F 10.8 Tolerances for holes and inserts

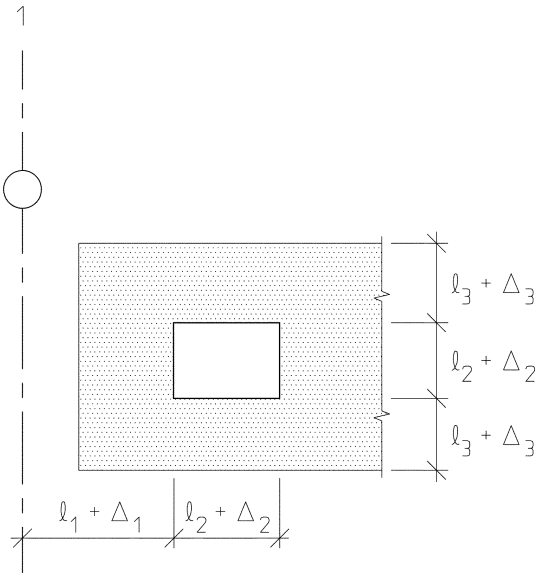
No.	Type of deviation	Description	Permitted deviation △ Class 1
	 1 Reference line	$\Delta_1, \Delta_2, \Delta_3$	± 25 mm Unless otherwise stated in the project specification

Figure F.6 – Permitted deviations for holes and inserts

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Annex G (informative): Guidance on inspection

Main clause numbers mirror those in Section 11: Inspection

G 11.1 Inspection classes

Table G.1 gives guidance for selection of inspection classes.

The inspection class to be applied should be according to provisions valid at the construction site, and be stated in the project specification

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Table G.1 – Guidance for the selection of inspection class

Subject	Inspection Class 1	Inspection Class 2	Inspection Class 3
Type of construction works	- Buildings ≤ 2 storeys	- Ordinary bridges - Buildings > 2 storeys	- Special bridges - High rise buildings - Large dams - Buildings for nuclear reactors - Containment structures
Type of structural components	- Reinforced beams and slabs with spans < 10 m - Simple walls and columns - Simple foundation structures	- Reinforced beams and slabs with spans > 10 m - Slender walls and columns - Pile caps - Arches < 10 m	- Reinforced arches and vaults - Highly compressed components - Very sensitive and complicated foundations - Arches > 10 m
Type of construction materials/technologies used Concrete acc. to prEN 206:1997: - Strength class - Exposure class Reinforcement	- Structures with precast elements Up to and incl. C25/30 X0, XC1, XC2, XA1, XF1 Ordinary	- Structures with precast elements Any strength class Any exposure class Ordinary and prestressing	- Structures with precast elements - Special tolerances Any strength class Any exposure class Ordinary and prestressing

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G 11.6 Inspection of prestressing

G.11.6.2 Inspection prior to concreting

(1) In addition to the requirements in 11.6.2. the following items should be inspected:

– that the anchor plates are perpendicular to the tendons;
– straight alignment of the tendons in the region of anchors and couplings;
– check to ensure an adequate length of prestressing steel for the installation of the jacks.

G 11.7 Inspection of the concreting operations

(1) A detailed inspection plan, where required by the project specification, should identify all inspection, monitoring and testing necessary to prove that the required quality has been achieved.

An inspection plan should, for each inspection point state:

– the requirements;
– the references to the standard and the project specification;
– the method of inspection, monitoring or testing;
– the definition of inspection section;
– the frequency of inspection, monitoring or testing;
– the acceptance criteria;
– the documentation;
– the responsible inspector;
– the owner's witness points, if any.

An inspection plan may be prepared as a summary table with references to the inspection procedures and inspection instructions giving the details of inspection, monitoring and testing.

All forms to be used for documentation should be accepted by the owner or his representative before the construction starts.

(2) Tables G.2 to G.7 give guidance on inspection of concreting operations.

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Table G.2 – Inspection of pre-concreting and production

Subject	Method	Requirement	Inspection Class 1	Inspection Class 2	Inspection Class 3
Specification of concrete	Visual	prEN 206:1997	Before start of the production	Before start of the production	Before start of the production
Inspection of concrete production	Examination of certificate where available Visual inspection where there is no 3rd party inspection	Certificate from approved certification body that production is controlled (according to prEN 206:1997) Otherwise inspection of production plant (according to prEN 206:1997)	New supplier and in case of doubt New supplier and in case of doubt	New supplier and in case of doubt New supplier and in case of doubt	New supplier and in case of doubt New supplier and in case of doubt
Planning for production	Visual inspection	Relevant information for production		Written information	Written information

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Table G.3 – Inspection of fresh concrete

Subject	Method	Requirement	Inspection Class 1	Inspection Class 2	Inspection Class 3
Delivery ticket where relevant	Visual inspection	Conformity to the specification	Each delivery	Each delivery	Each delivery
Consistence of concrete	Visual inspection Using an appropriate consistence test [1)]	Consistence as ordered Conformity to consistence class	Random Only when in doubt	Each delivery When making test for hardened concrete and when in doubt	Each delivery When making test for hardened concrete and when in doubt
Uniformity of concrete	Visual inspection Test by comparing the properties of sub samples taken from different part of a batch [3)]	Homogenous concrete appearance Sub samples shall show the same properties[4)]	 When in doubt	Each delivery When in doubt	Each delivery When in doubt
Identity testing for compressive strength	Test according to prEN 206:1997 [2)]	Conformity to compressive strength class[2)]	For concrete without CE-marking or other third party certification When in doubt	For concrete without CE-marking or other third party certification In accordance with project specification When in doubt	For concrete without CE-marking or other third party certification In accordance with project specification When in doubt
Air content	On-site-test according to prEN 206:1997 [1)]	Conformity to specification	Random In accordance with project specification When in doubt	Random In accordance with project specification When in doubt	In accordance with project specification When in doubt
Other characteristics: Retempering Time of arrival Time of placing Temperature	[3)] Record Record Record Record	[3)] Dosage and type of agent [5)] [5)] [5)]	 Each delivery to be recorded When required When required When required	 Each delivery to be recorded When required When required When required	 Each delivery to be recorded When required When required When required

NOTE:

[1)] The identity test shall be the criterion given in prEN 206:1997 for an individual sample.

[2)] Identity testing for strength, where required, ie concretes without CE-marking or third party certification.

[3)] In accordance with specified standards or to be agreed upon.

[4)] Within the precision of the test and agreed variability tolerances.

[5)] According to prEN 206:1997 and project specification

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Table G.4 – Inspection of pre-concreting operations

Subject	Inspection Class 1	Inspection Class 2	Inspection Class 3
Planning for inspection		Trial casting results, if any. Agreement on Quality Control. Inspection plan Equipment list	Trial casting results, if any. Agreement on Quality Control. Inspection plan Equipment list Operator list
Inspection	Basic inspection Inspection when in doubt	Basic and random inspection Stability of the falsework and formwork Visual inspection of: - tie bars - tightness of form - cleanliness of form - release agent, amount - saturation of form - construction joint -casting sequence planned - access - delivery planned – concrete cover Measuring dimensions	Inspection before each casting Stability of the falsework and formwork Visual inspection of: - tie bars - tightness of form - cleanliness of form - release agent, amount - saturation of form - construction joint - casting sequence planned - access - delivery planned – concrete cover Measuring dimensions

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Table G.5 – Inspection of placing and compaction

Subject	Inspection Class 1	Inspection Class 2	Inspection Class 3
Planning for inspection		Instruction to operators Rate of placing Placing sequence Layer thickness	Instruction to operators Rate of placing Placing sequence Layer thickness Drawing or process diagram
Inspection, formed surfaces	Basic inspection	Basic and random inspection: - weather conditions - rate of placing - placing sequence - layer thickness - segregation - consistence - number of pokers - size of pokers - intrusion distance - intrusion depth - re-vibration - form vibrators - surface vibrators - movements of concrete - deflection of form - fixation of embedded parts	Inspection entire casting: - weather conditions - rate of placing - placing sequence - layer thickness - segregation - consistence - number of pokers - size of pokers - intrusion distance - intrusion depth - re-vibration - form vibrators - surface vibrators - movements of concrete - deflection of form - fixation of embedded parts
Inspection, free surfaces	Basic inspection	Basic and random inspection: - laitance on top - evenness of surface - formation of crust - time end of compaction - time of completion - protection of surface Measuring of surface deviations in accordance with project specification	Inspect entire casting: - laitance on top - evenness of surface - formation of crust - time end of compaction - time of completion - protection of surface Measuring of surface deviations in accordance with project specification

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Table G.6 – Inspection of curing and protection

Subject	Inspection Class 1	Inspection Class 2	Inspection Class 3
Planning for inspection		Procedure for protection against drying out and freezing Procedure for temperature control Monitoring system for temperature and maturity registration	Procedure for protection against drying out and freezing Procedure for temperature control Monitoring system for temperature and maturity registration Calculation of temperature development and distribution in accordance with project specification
Inspection	Basic inspection	Basic and random inspection: - protection against drying out, maturity - protection against freezing - stripping time, maturity - temperature differences	Inspection each casting: - protection against drying out, maturity - protection against freezing - stripping time, maturity - temperature differences

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Table G.7 – Inspection of post-concreting operations

Subject	Inspection Class 1	Inspection Class 2	Inspection Class 3
Planning for inspection		Instruction for inspection in accordance with project specification	
Inspection	Geometrical check Basic inspection	Geometrical check Strength and maturity at time of stripping Surface appearance: - holes - honey combing - sandstripes - blow holes - cracks - crack widths Connections: - starter bars - bolts - inserts - fixtures Cover: - check with covermeter if required by project specification	

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G 11.8 Inspection of precast concrete elements

G 11.8.1 Initial inspection

The principal provisions to be verified by an initial inspection of the site, before installation of precast elements, are:

- the access routes for elements and equipment;
- the availability of assistance from the main contractor;
- the availability of adequate lifting equipment;
- the availability of proper equipment for safe working;
- the adequate completion of the supporting structures;
- provisional works, such as propping, scaffoldings, temporary supports, as necessary;
- as-built report documenting any deviation of the site work related to the erection.

G 11.8.2 Reception checks

(1) The requirements for the acceptance inspection of precast concrete elements on site is given in table G.8

Table G.8 – Reception inspection of precast concrete elements

Subject	Property	Method	Frequency	Action
elements	marking, quantity	visual inspection	every element	signature on the delivery ticket and notice of imperfections
elements	evident imperfections	visual inspection	every element	signature on the delivery ticket and notice of imperfections
elements	appearance of joint faces	visual inspection	every element	signature on the delivery ticket and notice of imperfections
lifting devices in the element	type, integrity and compatibility	visual inspection	every element	signature on the delivery ticket and notice of imperfections

G 11.8.3 Additional checks

(1) When appropriate, the reception inspection should include the items in table G.9.

Table G.9 – Additional items for inspection when appropriate

Subject	Property	Method	Frequency	Action
elements	geometrical tolerances	standard test methods	in case of doubt	full report
elements	crack width and extension	microscope and tape/rule	if required	full report
elements	joint shapes and dimensions	tape/rule	in case of doubt	full report
elements	other characteristics	standard test methods	standard test methods	full report

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G 11.9 Action in the event of a non-conformity

(1) Where non-conformity is confirmed, the following aspects should be investigated in the listed order:

– the implications of the non-conformity on execution and service;
– the measures necessary to make the component acceptable;
– the necessity of rejection and replacement of the non-repairable component.

NOTE If the implications of the non-conformity on execution and service are negligible, the component should be accepted. If the non-conformity can be corrected, the component should be accepted after proper repair.

(2) If required, the rectification of non-conformity should be in accordance with a procedure stated in the project specification or agreed.

Documentation of the procedure and materials to be used should be approved before the corrections are made.

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